

de Laval nozzle

A **de Laval nozzle** (or **convergent-divergent nozzle**, **CD nozzle** or **con-di nozzle**) is a tube which is pinched in the middle, making a carefully balanced, asymmetric hourglass shape. It is used to accelerate a compressible fluid to supersonic speeds in the axial (thrust) direction, by converting the thermal energy of the flow into kinetic energy. De Laval nozzles are widely used in some types of steam turbines and rocket engine nozzles. It also sees use in supersonic jet engines.

Similar flow properties have been applied to jet streams within astrophysics.^[1]

History

Giovanni Battista Venturi designed converging-diverging tubes known as Venturi tubes to experiment the effects in fluid pressure reduction while flowing through chokes (Venturi effect). German engineer and inventor Ernst Körting supposedly switched to a converging-diverging nozzle in his steam jet pumps by 1878 after using convergent nozzles but these nozzles remained a company secret.^[2] Later, Swedish engineer Gustaf de Laval applied his own converging diverging nozzle design for use on his impulse turbine in the year 1888.^{[3][4][5][6]}

Laval's convergent-divergent nozzle was first applied in a rocket engine by Robert Goddard. Most modern rocket engines that employ hot gas combustion use de Laval nozzles.

Operation

Its operation relies on the different properties of gases flowing at subsonic, sonic, and supersonic speeds. The speed of a subsonic flow of gas will increase if the pipe carrying it narrows because the mass flow rate is constant. The gas flow through a de Laval nozzle is isentropic (gas entropy is nearly constant). In a subsonic flow, sound will propagate through the gas. At the "throat", where the cross-sectional area is at its minimum, the gas velocity locally becomes sonic (Mach number = 1.0), a condition called choked flow. As the nozzle cross-sectional area increases, the gas begins to

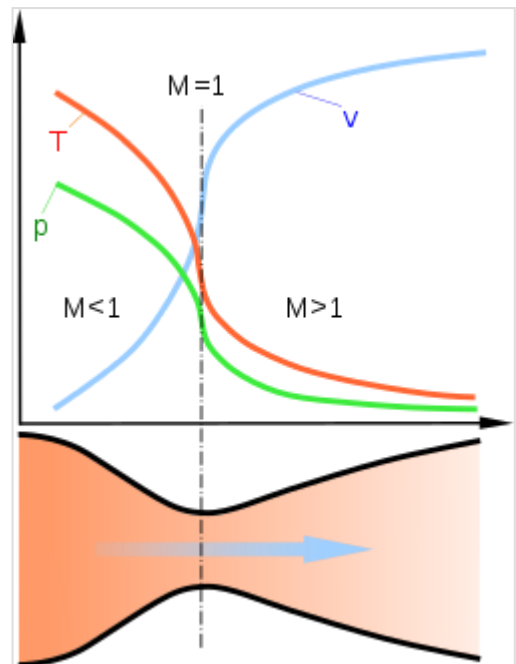


Diagram of a de Laval nozzle, showing approximate flow velocity (v), together with the effect on temperature (T) and pressure (p)



Longitudinal section of RD-107 rocket engine (Tsiolkovsky State Museum of the History of Cosmonautics)

expand, and the flow increases to supersonic velocities, where a sound wave will not propagate backward through the gas as viewed in the frame of reference of the nozzle (Mach number > 1.0).

As the gas exits the throat, the increase in area allows it to undergo a Joule–Thomson expansion, wherein the gas expands at supersonic speeds from high to low pressure, pushing the velocity of the mass flow beyond sonic speed.

When comparing the general geometric shape of the nozzle between the rocket and the jet engine, it only looks different at first glance, when in fact is about the same essential facts are noticeable on the same geometric cross-sections – that the combustion chamber in the jet engine must have the same "throat" (narrowing) in the direction of the outlet of the gas jet, so that the turbine wheel of the first stage of the jet turbine is always positioned immediately behind that narrowing, while any on the further stages of the turbine are located at the larger outlet cross-section of the nozzle, where the flow accelerates.

Conditions for operation

A de Laval nozzle will only choke at the throat if the pressure and mass flow through the nozzle is sufficient to reach sonic speeds, otherwise no supersonic flow is achieved, and it will act as a Venturi tube; this requires the entry pressure to the nozzle to be significantly above ambient at all times (equivalently, the stagnation pressure of the jet must be above ambient).

In addition, the pressure of the gas at the exit of the expansion portion of the exhaust of a nozzle must not be too low. Because pressure cannot travel upstream through the supersonic flow, the exit pressure can be significantly below the ambient pressure into which it exhausts, but if it is too far below ambient, then the flow will cease to be supersonic, or the flow will separate within the expansion portion of the nozzle, forming an unstable jet that may "flop" around within the nozzle, producing a lateral thrust and possibly damaging it.

In practice, ambient pressure must be no higher than roughly 2–3 times the pressure in the supersonic gas at the exit for supersonic flow to leave the nozzle.

Analysis of gas flow in de Laval nozzles

The analysis of gas flow through de Laval nozzles involves a number of concepts and assumptions:

- For simplicity, the gas is assumed to be an ideal gas.
- The gas flow is isentropic (i.e., at constant entropy). As a result, the flow is reversible (frictionless and no dissipative losses), and adiabatic (i.e., no heat enters or leaves the system).
- The gas flow is constant (i.e., in steady state) during the period of the propellant burn.
- The gas flow is along a straight line from gas inlet to exhaust gas exit (i.e., along the nozzle's axis of symmetry)
- The gas flow behaviour is compressible since the flow is at very high velocities (Mach number > 0.3).

Exhaust gas velocity

As the gas enters a nozzle, it is moving at subsonic velocities. As the cross-sectional area contracts the gas is forced to accelerate until the axial velocity becomes sonic at the nozzle throat, where the cross-sectional area is the smallest. From the throat the cross-sectional area then increases, allowing the gas to expand and

the axial velocity to become progressively more supersonic.

The linear velocity of the exiting exhaust gases can be calculated using the following equation:^{[7][8][9]}

$$v_e = \sqrt{\frac{TR}{M} \cdot \frac{2\gamma}{\gamma-1} \cdot \left[1 - \left(\frac{p_e}{p}\right)^{\frac{\gamma-1}{\gamma}}\right]},$$

where:

v_e = exhaust velocity at nozzle exit,

T = absolute temperature of inlet gas,

R = universal gas law constant,

M = the gas molecular mass (also known as the molecular weight)

$\gamma = \frac{c_p}{c_v}$ = isentropic expansion factor

(c_p and c_v are specific heats of the gas at constant pressure and constant volume respectively),

p_e = absolute pressure of exhaust gas at nozzle exit,

p = absolute pressure of inlet gas.

Some typical values of the exhaust gas velocity v_e for rocket engines burning various propellants are:

- 1,700 to 2,900 m/s (3,800 to 6,500 mph) for liquid monopropellants,
- 2,900 to 4,500 m/s (6,500 to 10,100 mph) for liquid bipropellants,
- 2,100 to 3,200 m/s (4,700 to 7,200 mph) for solid propellants.

As a note of interest, v_e is sometimes referred to as the *ideal exhaust gas velocity* because it is based on the assumption that the exhaust gas behaves as an ideal gas.

As an example calculation using the above equation, assume that the propellant combustion gases are: at an absolute pressure entering the nozzle $p = 7.0$ MPa and exit the rocket exhaust at an absolute pressure $p_e = 0.1$ MPa; at an absolute temperature of $T = 3500$ K; with an isentropic expansion factor $\gamma = 1.22$ and a molar mass $M = 22$ kg/kmol. Using those values in the above equation yields an exhaust velocity $v_e = 2802$ m/s, or 2.80 km/s, which is consistent with above typical values.

Technical literature often interchanges without note the universal gas law constant R , which applies to any ideal gas, with the gas law constant R_s , which only applies to a specific individual gas of molar mass M . The relationship between the two constants is $R_s = R/M$.

Mass flow rate

In accordance with conservation of mass the mass flow rate of the gas throughout the nozzle is the same regardless of the cross-sectional area.^[10]

$$\dot{m} = \frac{Ap_t}{\sqrt{T_t}} \cdot \sqrt{\frac{\gamma M}{R}} \cdot \text{Ma} \cdot \left(1 + \frac{\gamma-1}{2} \text{Ma}^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

where:

$\dot{m}$ = mass flow rate,

A = cross-sectional area ,

p_t = total pressure,

T_t = total temperature,

$\gamma = \frac{c_p}{c_v}$ = isentropic expansion factor,

R = universal gas constant,

Ma = Mach number

M = the gas molecular mass (also known as the molecular weight)

When the throat is at sonic speed $Ma = 1$ where the equation simplifies to:

$$\dot{m} = \frac{Ap_t}{\sqrt{T_t}} \cdot \sqrt{\frac{\gamma M}{R}} \cdot \left(\frac{\gamma + 1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}}$$

By Newton's third law of motion the mass flow rate can be used to determine the force exerted by the expelled gas by:

$$F = \dot{m} \cdot v_e$$

where:

F = force exerted,

$\dot{m}$ = mass flow rate,

v_e = exit velocity at nozzle exit

In aerodynamics, the force exerted by the nozzle is defined as the thrust.

See also

- History of the internal combustion engine
- Spacecraft propulsion
- Twister supersonic separator
- Isentropic nozzle flow
- Daniel Bernoulli

References

1. C.J. Clarke and B. Carswell (2007). *Principles of Astrophysical Fluid Dynamics* (https://archive.org/details/principlesastrop00clar_804) (1st ed.). Cambridge University Press. pp. 226 (https://archive.org/details/principlesastrop00clar_804/page/n237). ISBN 978-0-521-85331-6.

2. Krehl, Peter O. K. (24 September 2008). *History of Shock Waves, Explosions and Impact: A Chronological and Biographical Reference* (<https://books.google.com/books?id=PmuqCHDC3pwC&q=nozzle+Ernst+Koerting&pg=PA396>). Springer. ISBN 9783540304210. Archived (<https://web.archive.org/web/20210910180759/https://books.google.com/books?id=PmuqCHDC3pwC&q=nozzle+Ernst+Koerting&pg=PA396>) from the original on 10 September 2021. Retrieved 10 September 2021.
3. See:
 - Belgian patent no. 83,196 (issued: 1888 September 29)
 - English patent no. 7143 (issued: 1889 April 29)
 - de Laval, Carl Gustaf Patrik, "Steam turbine," (<http://pdfpiw.uspto.gov/piw?docid=00522066&PageNum=1&IDKey=881F85454D87>) Archived (<https://web.archive.org/web/20180111165305/http://pdfpiw.uspto.gov/piw?docid=00522066&PageNum=1&IDKey=881F85454D87>) 2018-01-11 at the Wayback Machine U.S. Patent no. 522,066 (filed: 1889 May 1; issued: 1894 June 26)
4. Theodore Stevens and Henry M. Hobart (1906). *Steam Turbine Engineering*. MacMillan Company. pp. 24–27. Available on-line here (<https://books.google.com/books?id=9EIMAAAMAAJ&pg=PA27>) Archived (<https://web.archive.org/web/20141019162649/http://books.google.com/books?id=9EIMAAAMAAJ&pg=PA27&lpg=PA26&ots=i9N3YYNjIF&ie=ISO-8859-1&output=html>) 2014-10-19 at the Wayback Machine in Google Books.
5. Robert M. Neilson (1903). *The Steam Turbine* (<https://archive.org/details/steamturbine03neilgoog>). Longmans, Green, and Company. pp. 102 (<https://archive.org/details/steamturbine03neilgoog/page/n126>)–103. Available on-line here (<https://archive.org/details/steamturbine07neilgoog/page/n128>) in Google Books.
6. Garrett Scaife (2000). *From Galaxies to Turbines: Science, Technology, and the Parsons Family*. Taylor & Francis Group. p. 197. Available on-line here (<https://books.google.com/books?id=BeMjgxsifcQC&pg=PA197>) Archived (https://web.archive.org/web/20141019161848/http://books.google.com/books?id=BeMjgxsifcQC&pg=PA197&lpg=PA197&source=bl&ots=VpRffcaLG2&sig=mNb8dGDFErN8mgmo79HN6Dpa2DM&hl=en&ei=jMkJS82WFJHIIAew4IH9CQ&sa=X&oi=book_result&ct=result&resnum=10&ved=0CCUQ6AEwCTgU) 2014-10-19 at the Wayback Machine in Google Books.
7. "Richard Nakka's Equation 12" (http://www.nakka-rocketry.net/th_nozz.html). Archived (https://web.archive.org/web/20170715044159/http://www.nakka-rocketry.net/th_nozz.html) from the original on 2017-07-15. Retrieved 2008-01-14.
8. "Robert Braeuning's Equation 1.22" (<http://www.braeunig.us/space/propuls.htm#intro>). Archived (<https://web.archive.org/web/20060612212910/http://www.braeunig.us/space/propuls.htm#intro>) from the original on 2006-06-12. Retrieved 2006-04-15.
9. George P. Sutton (1992). *Rocket Propulsion Elements: An Introduction to the Engineering of Rockets* (6th ed.). Wiley-Interscience. p. 636. ISBN 0-471-52938-9.
10. Hall, Nancy. "Mass Flow Choking" (<https://www.grc.nasa.gov/WWW/K-12/airplane/mflchk.html>). NASA. Archived (<https://web.archive.org/web/20200808132353/https://www.grc.nasa.gov/WWW/K-12/airplane/mflchk.html>) from the original on 8 August 2020. Retrieved 29 May 2020.

External links

- Exhaust gas velocity calculator (<https://www.fxsolver.com/browse/formulas/Exhaust+Gas+Velocity>)
- Other applications of nozzle theory Flow of gases and steam through nozzles (<https://fluid-dynamics.education/flow-of-gases-and-steam-through-nozzles.html>)

Retrieved from "https://en.wikipedia.org/w/index.php?title=De_Laval_nozzle&oldid=1196339499"

■